

Influence of Turbidity on Brook Trout Reactive Distance and Foraging Success

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Abstract.—Past research has focused on the effects of sediment action on stream morphology and the habitat of brook trout *Salvelinus fontinalis*. Throughout the Appalachian Mountains, the watersheds in which brook trout reside are being influenced by timber harvesting and related road construction. Although these streams may have gradients steep enough to prevent deleterious sediment deposition, elevated stream turbidity is nevertheless common. An understanding of the sublethal effects of increased sedimentation and turbidity is essential to further our knowledge of the effects of increased sediment loading on stream fish production and how these effects differ among species living in sympatry. The specific objectives of this study were to determine (1) the effects of turbidity on the reactive distance of brook trout, (2) how turbidity affects encounter rates between brook trout and their prey, and (3) how turbidity affects brook trout's foraging success. We used videographic techniques to study brook trout foraging behavior in an artificial stream. Three brook trout were tested during each sampling period, creating a competitive situation under which a more accurate measure of reactive distance could be made. Treatment turbidity levels ranged from 0 to 43 nephelometric turbidity units. The reactive distance of brook trout decreased curvilinearly with increasing turbidity. The probability of a brook trout's reacting to a given prey item was correlated with reactive distance and also decreased with turbidity. However, turbidity had no influence on the probability of attack given a reaction, the probability of capture given an attack, or the probability of ingestion given capture. In natural streams, invertebrate drift densities typically increase with turbidity as flows increase, but this increase may not be enough to compensate for the decreased ability of brook trout to detect drifting prey.

Increased fine-sediment loading can negatively influence fish production owing to the reduced availability and permeability of spawning gravel (Davies and Nelson 1993; Hartman et al. 1996), the reduction of invertebrate density (Hartman et al. 1996), and altered stream geomorphology (Alexander and Hansen 1986). Whereas bed sediments affect fishes by altering substrate composition and reproductive potential, suspended sediments influence organisms by reducing light penetration and visual clarity (Barret et al. 1992), increasing physiological stress, reducing tolerance to disease (Redding et al. 1987), and causing respiratory impairment (Berg and Northcote 1985). However, turbidity seldom reaches lethal levels in natural systems. This suggests that the effects of suspended sediment on community structure are largely the products of behavioral responses (Graddall and Swenson 1982). Relatively high sediment loads may be required to alter stream morphology, but low inputs can raise the amount of suspended sediment and markedly increase turbidity (Duchrow and Everhart 1971). Turbidity may have

greater influence on fish production than any other sediment-related effect (Barret et al. 1992).

Responses of individual fish to turbidity vary. Steelhead (anadromous rainbow trout) *Oncorhynchus mykiss* and coho salmon *O. kisutch* avoid turbid water (Bisson and Bilby 1982; Sigler et al. 1984). Avoidance behavior by salmonids can result in the absence of salmonids in long stream reaches (Waters 1995), possibly in an effort to maintain view of potential prey (Bisson and Bilby 1982). While some researchers have found that elevated turbidity levels can reduce the feeding and growth rates of fishes (Sigler et al. 1984; Berg and Northcote 1985; McLeay et al. 1987; Reynolds et al. 1989), others have reached different conclusions (Swenson and Matson 1976; Breitung 1988; Gregory and Northcote 1993).

Reduced growth under turbid conditions may result from reductions in reactive distance and prey encounter rate as much as from reductions in prey productivity. Although the reactive distance of a fish has been defined in various ways in the literature (Barret et al. 1992; Benfield and Minello 1996), the general consensus is that as turbidity increases, a fish's reactive distance will decrease owing to inhibited visual clarity (Gardner 1981; Sigler et al. 1984; Barret et al. 1992; Miner and

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Stein 1996). The reactive distance of juvenile bluegills *Lepomis macrochirus* to prey was reduced to less than 3 cm at turbidities greater than 60 nephelometric turbidity units (NTU; Gardner 1981). Their reactive distance to piscivorous largemouth bass *Micropterus salmoides* also decreased as a negative power function of increasing turbidity (Miner and Stein 1996). In contrast, Barret et al. (1992) found a negative linear relationship between the reactive distance of rainbow trout and turbidity, with a 55% reduction in reactive distance at 30 NTU compared with clear water. Mobile prey could escape in these reduced fields of vision, limiting consumption and growth. Turbidity could also alter the feeding selectivity of fish, causing them to capture only slowly moving and slowly reacting prey types (Gardner 1981).

There has been considerable research on the effects of turbidity on salmonid foraging (Bisson and Bilby 1982; Sigler et al. 1984; Barret et al. 1992), yet studies with brook trout *Salvelinus fontinalis* are lacking. Brook trout are an important component of fish communities in Appalachian streams, often being the only species present. Their habitat is easily degraded by land-use practices such as timber harvesting and road construction. Turbidity may thus be an important environmental feature contributing to the diminution of brook trout production.

We conducted an experimental study to determine foraging behavioral responses of brook trout to turbidity. We addressed the following questions: (1) how turbidity affects the reactive distance of brook trout, (2) how turbidity affects encounter rates between brook trout and their prey, and (3) how turbidity affects the foraging success of brook trout.

Methods

We conducted our experiments in an artificial stream. All feeding trials were videotaped and then analyzed to determine reactive distance. Foraging success was determined by observing the brook trout on a computer monitor while trials were being videotaped to determine if each prey item elicited a reaction, was attacked, was captured, and was ingested.

Artificial stream design.—The artificial stream used to measure reactive distance was a galvanized steel tank measuring 1.2×2.4 m. The tank had two removable 1.8-m baffles that separated it into one large and two small longitudinal sections. The largest section was used to observe the fish. Fish were contained by fences of 1-cm²-square galva-

nized wire fencing material positioned upstream and downstream of the viewing portion of the tank. The artificial stream was surrounded by a curtain to minimize outside disturbance. Current (0.06 m/s) within the artificial stream was created with the use of three pumps placed in front of the viewing area. A 20-cm piece of bright yellow poster board was placed on one of the cross supports of the artificial stream. This served as a reference target on which measures of reactive distance were based (see the section on reactive distance estimation below). The artificial stream was illuminated by a 150-W incandescent light reflected off the ceiling of the laboratory. Light intensity at the water's surface was 49 lux. Water temperatures were maintained by using a chiller in the head tank. Temperature was measured during each trial and averaged 12.2°C (range, 11.0–13.8°C).

Fish.—Brook trout were obtained from Bowden Fish Hatchery, Bowden, West Virginia. Mean ($\pm$ 95% confidence interval [CI]) total fish length was 136 ± 11.48 mm and mean ($\pm$ 95% CI) weight was 25.35 ± 6.65 g. We initially tried to use wild brook trout but found that they would not feed readily in the laboratory. Therefore, we decided to test members of the species that fed well on the offered prey.

Our experimental protocol for reactive distance measurement differed slightly from others reported in the literature. In our experiment, three brook trout were tested at a time, whereas other studies have measured reactive distance using only a single fish (e.g., Vinyard and O'Brien 1976; Barret et al. 1992). Previous reactive-distance work using a single fish assumed that the fish first saw the prey (or predator) at the same time the observer detected behavior in the fish indicating this sighting. However, the fish could have seen the prey long before the observer saw any signs of detection, especially in a flowing environment. If so, the true reactive distance would be underestimated. With three fish in a competitive trial, the individual that reacted to a potential prey item first would have an advantage over the other two. Thus, the use of more than one fish in reactive-distance experiments could reduce the time between detection of the prey and the observer's detection of behavioral changes in the fish. In this way, a better measure of reactive distance would be obtained.

To begin each experiment, brook trout were anesthetized with tricaine methanesulfonate (MS-222), weighed, measured, and placed into the viewing portion of the artificial stream. Each group of three fish was given 12 h to acclimate to the

stream before testing began. A total of seven groups of three fish were tested.

Videotaping of foraging behavior.—Recording of reactive distance and foraging behavior was conducted with a video camera mounted on the ceiling of the laboratory, 2 m above the viewing area of the artificial stream. The camera was connected to a computer, and the image of the tank and fish could be viewed on the monitor during videotaping of foraging trials.

Treatments.—Treatment turbidity levels were created using soil obtained from the banks of Roaring Creek, a native brook trout stream in Preston County, West Virginia. This soil was largely composed of clay particles. The soil was suspended by agitation in a bucket, creating a turbid solution. This solution was then added to the head tank until a desired turbidity treatment level was reached in the artificial stream.

Each group of three fish was tested at sequentially higher levels of turbidity. We defined a trial as one testing situation with a group of three fish at a single level of turbidity. A set of trials consisted of all the trials in which a group of three fish was tested, ranging from clear water (0 NTU) to the highest treatment level (approximately 40 NTU).

Water depth within the artificial stream was either 15 or 25 cm. In the first five sets of trials, the depth was 25 cm and turbidity treatment levels were approximately 0, 10, 15, 20, and 25 NTU. In 25 cm of water, it became nearly impossible to accurately view the fish at turbidities above 25 NTU. Water depth was then lowered to 15 cm, which enabled continued testing of fish up to approximately 40 NTU. Two additional sets of trials were conducted in 15 cm of water at turbidity levels of approximately 0, 10, 15, 20, 25, 30, 35, and 40 NTU.

Feeding trials were conducted with live larvae of houseflies *Musca domestica* as prey items. Mean ($\pm$ SE) length of the larvae was 10.3 ± 0.2 mm. These larvae were small enough that a fish could consume many without becoming satiated yet large enough to be seen with the camera. Because the larvae normally sank quickly, air was injected into each of them with a syringe so that all prey floated before a feeding trial began. After injection, the larvae were largely immobile, so that we could maintain our view of both predator and prey.

Feeding trials.—Each feeding trial lasted approximately 20 min, during which 25–30 housefly larvae were sequentially placed in the viewing area by rinsing them down a delivery tube. To prevent

habituation to a single location of prey offer, there were three such tubes at the front of the viewing portion of the artificial stream. After each prey item was released, it was observed and its fate recorded. A “reaction” was defined as any action a fish took that included orientating to a larva. Pursuit of a larva was considered an “attack.” If the fish took the larva into its mouth, that was considered a “capture.” Finally, if a fish did not reject a larva after capturing it, we considered the event an “ingestion.” Subsequent prey items were not offered until the previous one was either consumed or passed out of the back of the viewing area. Unconsumed prey were captured in a net at the rear of the viewing area.

With the light color of the artificial stream, there was not sufficient contrast to allow observation of the light-colored prey at low turbidities. To remedy this problem, a 0.8-m black plastic mat was placed in the front portion of the viewing area. The mat was used only at low turbidity levels (≤ 15 NTU); it was removed at higher levels so that the fish could be seen against a light background. Removal of the mat did not appear to influence fish behavior in any way. Because all prey floated on the surface and the fish only saw a contrast between the prey and the outside lighting (i.e., not with anything in the artificial stream), we feel that the presence or absence of the mat did not influence our results in any way.

At the end of a feeding trial, more-turbid water was added to the head tank until the next highest turbidity treatment level was reached. This process took approximately 10 min, after which the fish were again allowed to acclimate to the treatment for approximately 11.5 h before the next trial was conducted. Thus, two trials, each at a different treatment level, were conducted per day until the final trial was performed. At this time, the fish were removed. Individuals were only tested in a single series of turbidity levels (i.e., a set of trials). The artificial stream and head tank were then prepared for the next set of fish to be tested. We believe that testing the fish at sequential turbidity levels did not influence the results. In a similar study with rainbow trout, Barret et al. (1992) found that sequential application of treatment levels resulted in no difference in reactive distance compared to randomized treatments.

Reactive distance estimation.—We defined reactive distance as the distance between predator and prey at which the predator first showed any reaction to the prey. The feeding video was reviewed and frame-capture software used to capture

a video image at the instant of a fish's reaction to a prey item. At this time a red marker was placed on the fish's head and one on the prey item for measurement of the reactive distance.

Interactive Data Language software (IDL 5.0 from Research Systems) was used to measure reactive distance. Because the yellow reference target was closer to the camera lens than the objects in the viewing area, we corrected our distance estimates using the equation $\text{actual length} = 1.25 \times \text{estimated length} - 0.52$ ($R^2 = 0.98$) and actual lengths measured with stationary markers in the artificial stream. Mean differences ($\pm 95\%$ CI) between estimated and actual lengths were 0.52 ± 0.25 cm.

Statistical analysis.—Data were analyzed with the Windows-based SAS version 6.12. Our goal was to discover the model that best described how turbidity influenced brook trout reactive distance. Therefore, we only used the maximum recorded reactive distance in each trial in the analysis. Maximum reactive distances were $\log_e$ transformed. Because the same group of three fish was tested at all treatments in a given set of trials, we used a randomized-block analysis of variance (ANOVA) with each group of fish as blocks and performed polynomial regression to determine transformed parameters. Intercept parameters from the polynomial regression were exponentiated to arrive at the untransformed parameter estimates. This procedure was also used to determine the effect of depth on reactive distance, using the two values for depth as blocks.

As a means of evaluating the trade-offs between increased drift rate and reduced reactive distance at elevated turbidity in the field, we ran a simulation to determine the degree to which invertebrate drift would have to increase to compensate for reduced reactive distance. The Gerritsen and Strickler (1977) encounter rate model is given by the equation

$$E = (\pi R^2 N_H / 3) * (v^2 + 3u^2) / u$$

where E is the encounter rate in numbers per second, R is the predator's reactive distance, N_H is the prey density, v is the predator's swimming speed, and u is the prey's swimming speed. Because brook trout are drift feeders and capture prey by ambush, v was set to 0. The variable u was set to 0.06 m/s, which was the velocity of prey drifting in the artificial stream. Prey density, N_H , was set to that found within the viewing area of the tank while feeding, namely, 1.38 prey/m³. We then found the

mean encounter rate in clear water (< 3 NTU) for the brook trout in these experiments. Holding the mean encounter rate in clear water constant, we then solved the encounter rate model for the prey density necessary to maintain that encounter rate as reactive distance changed with increasing turbidity.

Foraging success data were converted to probabilities for each trial. The probability of reaction, $P(R)$, was equal to the number of prey eliciting a reaction divided by the number of prey offered. The probability of an attack given a reaction, $P(A|R)$, was set equal to the number of attacks divided by the number of reactions to a prey item. The probability of capture given an attack, $P(C|A)$, was set equal to the number of captures divided by the number of attacks. Finally, the probability of ingestion given capture, $P(I|C)$, was set equal to the number of ingested prey divided by the number of captured prey. These probabilities were arcsine-square-root transformed to more closely approximate a normal distribution and then analyzed in the same way as reactive-distance data.

Results

Reactive Distance

A total of 39 measurements of maximum reactive distance were included in the analysis. Four sets of trials with five treatment levels each and one set of trials with four treatment levels were conducted at a water depth of 25 cm. Two more sets of trials with eight treatment levels each were conducted at a water depth of 15 cm. One trial in clear water (0.57 NTU) was omitted from the data because the fish seemed lethargic and did not react readily to prey. This was the first trial in the sequence for this group of fish, and they may not have acclimated to the testing environment. In subsequent trials, they did not show signs of stress or lethargy.

Both turbidity and the group effect were significant. Turbidity decreased reactive distance significantly in both linear (ANOVA: $F_{1,30} = 437.98$; $P < 0.01$) and quadratic (ANOVA: $F_{1,30} = 9.86$; $P < 0.01$) model parameters. The group effect (our blocks in the ANOVA) was significant ($F_{6,30} = 7.48$; $P < 0.01$); the parameter estimates for each group were averaged and then added to the intercept parameter to obtain a mean intercept estimate for all groups. Water depth was not a significant factor (ANOVA: $F_{1,35} = 3.08$; $P = 0.09$). The model describing how turbidity influenced maximum brook trout reactive distance,

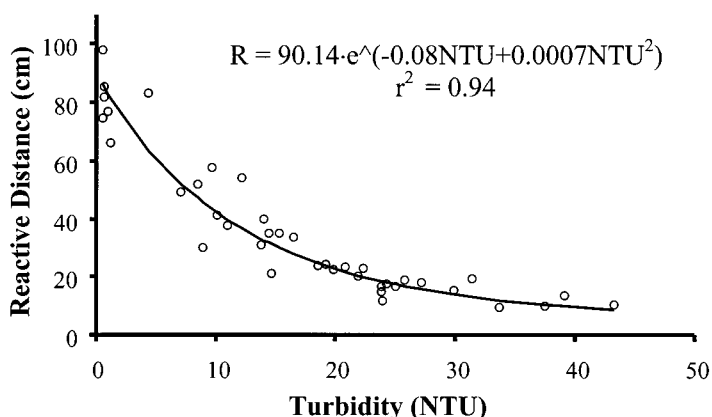


FIGURE 1.—Model describing the reactive distance of brook trout at turbidity levels ranging from 0 to 43 nephelometric turbidity units (NTU). Data from two depths (15 and 25 cm) and all turbidity levels were pooled for model estimation ($N = 39$).

$$R_{\max} = 90.14 \cdot \exp(-0.08 [\text{NTU}] + 0.00066 [\text{NTU}^2]),$$

fit the data well ($r^2 = 0.94$; Figure 1). Brook trout in clear water ($\text{NTU} < 3.0$) had a mean ($\pm 95\%$ CI) reactive distance of 80.43 ± 8.82 cm, which decreased to 12.38 ± 4.01 cm in the highest turbidities (> 30 NTU).

Encounter Rates

Reactive distance, R , is the highest-order variable in the Gerritsen and Strickler (1977) encounter rate model, and thus any changes in R produce the greatest relative changes in encounter rate. The mean ($\pm 95\%$ CI) encounter rate in clear water was 0.18 ± 0.02 prey/s. We then used our experimentally derived parameters for reactive distance in

the encounter rate model and solved for the density of prey needed to maintain the same mean encounter rate found in clear water over the range of turbidity levels in this study. According to our results, prey density must increase exponentially—and faster than reactive distance decreases—to maintain a constant encounter rate between brook trout and prey (Figure 2). For example, while reactive distance decreases by a factor of 2.1 in going from 0 to 10 NTU, prey density must increase by a factor of 4.5; and while reactive distance decreases by a factor of 4.0 in going from 0 to 20 NTU, prey density must increase by a factor of 16.

Foraging Success

As turbidity increased, the probability of reacting to an offered prey item decreased linearly (AN-

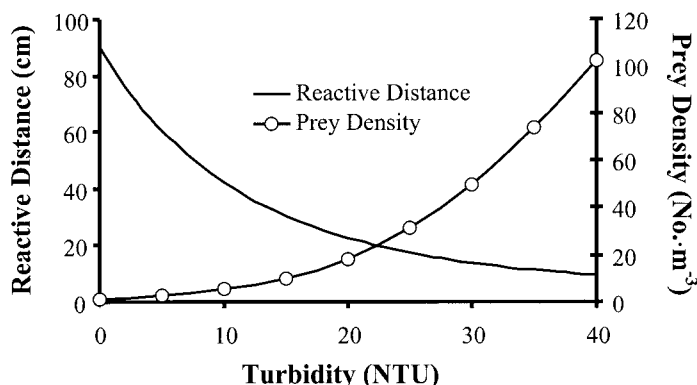


FIGURE 2.—Relationships between (1) turbidity and the reactive distance of brook trout and (2) turbidity and the prey density necessary to maintain a constant encounter rate between predator and prey if the predator does not increase swimming activity. Density is that needed to maintain a mean ($\pm 95\%$ confidence interval) encounter rate of 0.18 ± 0.02 prey/s, as calculated for brook trout in clear water (< 5 nephelometric turbidity units).

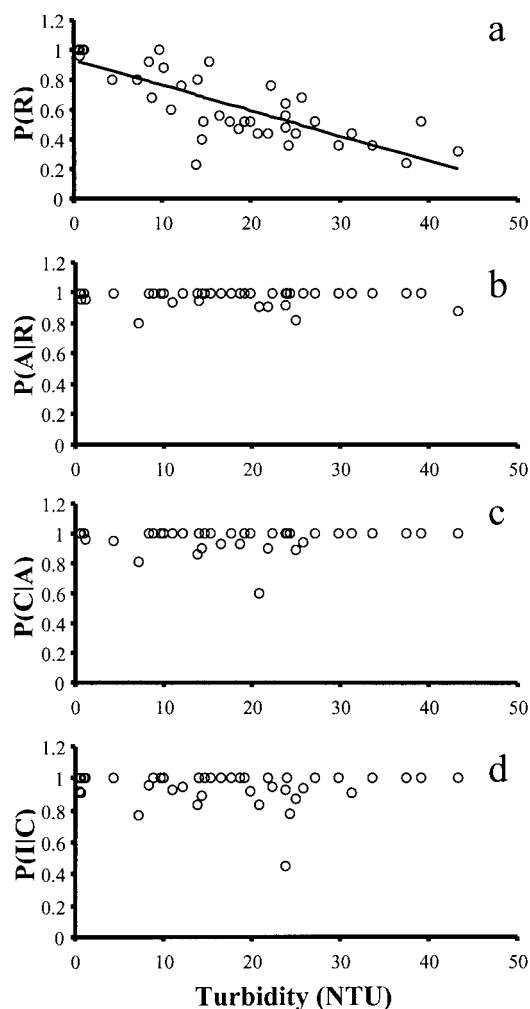


FIGURE 3.—Relationships between turbidity and (a) the probability of a brook trout's reacting to a prey item, (b) the probability of an attack given a reaction, (c) the probability of capture given an attack, and (d) the probability of ingestion given capture.

OVA: $F_{1,33} = 121.36$; $P < 0.01$) and there were no differences between groups of fish (ANOVA: $F_{6,33} = 0.01$; $P = 0.50$; Figure 3). We also found that the probability of a reaction was correlated with reactive distance (Pearson correlation coefficient = 0.85; $P < 0.01$). Thus, as reactive distance decreased, the probability of reacting to a given prey item also decreased. No differences were found in the probability of a reaction between water depths (ANOVA: $F_{1,38} = 0.01$; $P = 0.97$).

Turbidity had no influence on foraging success following a reaction to a prey item. The probability of an attack given a reaction (ANOVA: $F_{1,33} =$

2.07; $P = 0.16$), the probability of capture given an attack (ANOVA: $F_{1,33} = 1.10$; $P = 0.30$), and the probability of ingestion given capture (ANOVA: $F_{1,33} = 2.80$; $P = 0.10$) were not affected by increasing turbidity (Figure 3). Therefore, turbidity influenced detection of prey but not foraging success after a prey was detected.

Discussion

Reactive Distance

As expected, the reactive distance of brook trout decreased as turbidity increased and this decrease was best described by a nonlinear model. The curvilinear relationship between reactive distance and turbidity found in this study indicates that the greatest change in reactive distance per unit change in turbidity occurs at the lowest levels of turbidity (0–15 NTU). Other researchers have shown a nonlinear reduction in reactive distance with increasing turbidity for other species (Vinyard and O'Brien 1976; Gregory and Northcote 1993; Miner and Stein 1996). Benfield and Minello (1996) suggest that although reactive distance may differ between species, the shape of the relationship between reactive distance and turbidity is similar. In contrast, Barret et al. (1992) described the relationship as a linear function for rainbow trout.

A limitation of our reactive-distance results is that they only represent one prey type and one size range of predatory fish. Other factors, such as prey size and prey coloration or contrast with background light, can influence reactive distance. Vinyard and O'Brien (1976) found that the reactive distance of bluegills increases with prey size. Also, visual resolution is believed to increase as fish grow (Kao et al. 1985) and eye size increases (Huber and Rylander 1992). Our results should serve as a starting point for others doing reactive-distance work with brook trout and how turbidity relates to the foraging abilities and reactive distance of other species.

Encounter Rates

Encounter rates between predator and prey depend strongly on the reactive distance of the predator (Gerritsen and Strickler 1977). Slight changes in reactive distance resulted in pronounced effects on encounter rates and, in turbid waters, the Gerritsen and Strickler (1977) model would predict that encounter rates will decrease as turbidity increases. In theory, prey consumption should decrease in a similar manner as reactive distance with increasing turbidity. We found a reduction in the probability of reaction to a prey item with increas-

ing turbidity, but this was a linear relationship, not the curvilinear one that would be predicted by the encounter rate model. This is probably due to testing three fish at a time to determine reactive distance. The encounter rate model is applicable to a single predator, and the use of three fish in determining foraging success would increase the probability of an individual prey item's eliciting a reaction. However, our purpose here was not to test the Gerritsen and Strickler (1977) encounter rate model but to use this model to investigate how prey density would have to increase to compensate for decreased reactive distance.

For a brook trout to compensate for reduced encounter rates with prey, prey density must increase substantially or the fish must begin to actively search for prey. Salmonids have been shown to increase activity (Gradall and Swenson 1982) and feeding rates (Gregory and Northcote 1993) under turbid conditions. Increased activity has been attributed to decreased potential risk from predators (Gregory 1993; Gregory and Northcote 1993). Increased activity, however, reduces the net energy gain from each prey item consumed (Boisclair and Sirois 1993). Our encounter rate simulations show how prey density must increase to offset the reduction in reactive distance for salmonids that maintain a drift-feeding strategy and do not alter their activity in response to changes in turbidity. Invertebrate drift can increase under turbid conditions (Waters 1995), but it seems unlikely that it would increase to the degree necessary for brook trout to maintain the same encounter rate in turbid water as found in clear water without changes in activity. In a small stream in Oregon, Anderson and Lehmkuhl (1968) found that the total number of drifting invertebrates increased by a factor of four while discharge increased by a factor of five following approximately 2.5 cm of rainfall. Thus, although the total number of drifting invertebrates may increase under conditions that would bring about elevated turbidity levels, the drift density (number/[m³·s]) may remain relatively unchanged. Our encounter rate simulations predict that prey drift density must increase by a factor of 16 in waters of 20 NTU in order to maintain the same encounter rates found in clear water.

Foraging Success

The foraging success of brook trout was governed by the ability to detect and react to prey, but success after detection was not influenced by turbidity. As turbidity increased and reactive distance decreased, the probability of reaction also de-

creased. However, turbidity had no effect on the probability of attack given a reaction to prey, the probability of capture given an attack, or the probability of ingestion given capture.

It appears that reduced prey consumption by brook trout as a function of turbidity was governed primarily by a turbidity-induced decrease in prey detection. Several studies have found that other species show reduced consumption and slower growth rates in turbid water than in clear water (Birtwell et al. 1984; Sigler et al. 1984; Redding et al. 1987). In a study with coho salmon, Redding et al. (1987) used an index of intestinal fullness (0–3, with 3 representing a full intestine) to compare consumption across different turbidity levels. They found that after seven days' exposure to suspended sediment, mean indices were 2.8 for fish in clear water, 2.3 for fish in 0.3–0.6 mg/L suspended sediment, and 0.8 for fish in 2–3 mg/L suspended sediment. Sigler et al. (1984) found an approximately 75% decrease in growth rate for coho salmon and a 62% decrease in growth rate for steelhead reared in turbid water (49 NTU) compared with those reared in clear water. In reactive-distance experiments, Barret et al. (1992) also noted a decrease in the foraging strikes of rainbow trout at elevated turbidities. Contrary to these findings, Gregory and Northcote (1993) found the highest feeding rates of juvenile chinook salmon *Oncorhynchus tshawytscha* at turbidity levels from 35 to 150 NTU. Increased activity under turbid conditions (Gradall and Swenson 1982; Gregory 1993) may be responsible for these findings. This supports the concept that reactive distance alone may not be a good predictor of potential consumption and that behavioral changes should also be taken into account.

In natural streams, invertebrate drift has been found to increase with turbidity (Waters 1995). This may be the result of decreased light reaching the streambed and causing night-active diel drift photoperiodicity (Waters 1995), or it may be a response to settled suspended sediment (Ciborowski et al. 1977). Birtwell et al. (1984) found that invertebrate drift increased with increasing turbidity. However, Arctic grayling *Thymallus arcticus* in prey-rich turbid water had less in their stomachs than those in comparatively prey-poor clear water. Energy that fish might otherwise obtain from elevated invertebrate drift in turbid conditions might be damped by their inability to detect prey. Although short-term turbidity can increase invertebrate drift, streams with chronic turbidity can experience reduced primary (Lloyd et al. 1987) and

benthic invertebrate production over the long term, thus potentially further reducing the encounter rates between predator and prey.

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